

SOLIDWORKS API (.NET, VBA, C#) — Reference

(145/337): ILocalLinearPatternFeatureData

SOLIDWORKS API Documentation (converted from CHM)

ILocalLinearPatternFeatureData (Type)

SOLIDWORKS API Help

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ILocalLinearPatternFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ILocalLinearPatternFeatureData Interface

Allows access to a linear component pattern feature in an assembly.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ILocalLinearPatternFeatureData
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
```

C#

```
public interface ILocalLinearPatternFeatureData
```

C++/CLI

```
public interface class ILocalLinearPatternFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData](#).

Example

'VBA

'-----

' Preconditions: Verify that the assembly exists.

,

' Postconditions:

' 1. Opens the assembly.

' 2. Selects an edge for Direction 1.

' 3. Selects a subassembly to pattern.

' 4. Creates a local linear pattern.

' 5. Examine the FeatureManager design tree and

' graphics area.

,

' NOTE: Because the assembly is used elsewhere, do

' not save changes.

'-----

Option Explicit

Dim swApp As SldWorks.SldWorks

Dim swModel As SldWorks.ModelDoc2

Dim swModelDocExt As SldWorks.ModelDocExtension

Dim swFeatureManager As SldWorks.FeatureManager

Dim swFeature As SldWorks.Feature

Dim swLocalLinearPatternFD As SldWorks.LocalLinearPatternFeatureData

Dim status As Boolean

Dim errors As Long

Dim warnings As Long

Dim fileName As String

Sub main()

Set swApp = Application.SldWorks

fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2020\samples\tutorial\api\distance linkage.sldasm"

Set swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocASSEMBLY,
swOpenDocOptions_e.swOpenDocOptions_Silent, "", errors, warnings)

Set swModelDocExt = swModel.Extension

```
Set swFeatureManager = swModel.FeatureManager
```

```
status = swModelDocExt.SelectByID2("", "EDGE", 0.222723097227572, -0.193386735236572, -0.10172211746567, False, 2, Nothing, 0)
```

```
status = swModelDocExt.SelectByID2("mount base-1 @ distance linkage", "COMPONENT", 0, 0, 0, True, 1, Nothing, 0)
```

```
Set swLocalLinearPatternFD = swFeatureManager.CreateDefinition(swFmLocalLPattern)
```

```
swLocalLinearPatternFD.D1ReverseDirection = True  
swLocalLinearPatternFD.D1Spacing = 0.1516  
swLocalLinearPatternFD.D1TotalInstances = 4  
swLocalLinearPatternFD.D2PatternSeedOnly = False  
swLocalLinearPatternFD.D2ReverseDirection = False  
swLocalLinearPatternFD.D2Spacing = 0.05  
swLocalLinearPatternFD.D2TotalInstances = 1  
swLocalLinearPatternFD.SynchronizeFlexibleComponents = False
```

```
Set swFeature = swFeatureManager.CreateFeature(swLocalLinearPatternFD)
```

```
swModel.ClearSelection2 True  
swModel.ViewZoomtofit2
```

End Sub

Example

[Get and Set Seed Components \(C#\)](#)
[Get and Set Seed Components \(VB.NET\)](#)
[Get and Set Seed Components \(VBA\)](#)
[Create Local Linear Pattern \(C#\)](#)

Remarks

Read [Pattern Features and their Feature Data Objects](#).

Before calling [IFeatureManager::CreateDefinition](#), you must pre-select the D1 and D2 end (and seed) reference entities to create the linear component pattern feature.

If...	To select an up-to entity (vertex, edge, face, or plane), use...	To select a body, use...	To select a component, use...
Using IModelDocExtension::SelectByID2 to select entities, solid bodies, or components requires ordered selection	- Mark = 256 to select the Direction 1 end reference - Mark = 512 to select the Direction 1 end seed reference - Mark = 1024 to select the Direction 2 end reference - Mark = 2048 to select the Direction 2 end seed reference	- Mark = 1 to mark Direction 1 - Mark = 256 to mark the solid bodies to pattern	- Mark = 1 to mark the components to pattern - Mark = 2 to mark Direction 1 and Mark = 4 to mark Direction 2
Directly selecting an entity, solid body, or component	IEntity::Select4 with: ISelectData::Mark = 256 to select the Direction 1 end reference ISelectData::Mark = 512 to select the Direction 1 end seed reference ISelectData::Mark = 1024 to select the Direction 2 end reference ISelectData::Mark = 2048 to select the Direction 2 end seed reference	IEntity::Select4 with ISelectData::Mark = 1 to mark Direction 1 IBody2::Select2 with ISelectData::Mark = 256 to mark the bodies to pattern	IComponent2::Select4 and ISelectData::Mark = 1 to mark the components to pattern IEntity::Select4 and ISelectData::Mark = 2 for Direction 1; Mark=4 for Direction 2

After calling [IFeatureManager::CreateDefinition](#), you can further initialize the feature data object using:

- [ILocalLinearPatternFeatureData::D1Axis](#) to specify Direction 1.
- [ILocalLinearPatternFeatureData::SeedComponentArray](#) to specify the pattern's seed components.
- [ILocalLinearPatternFeatureData::D1EndCondition](#) to specify either spacing and instances or offsetting from an up-to reference:

If explicitly setting pattern spacing and number of pattern instances, use:

- [ILocalLinearPatternFeatureData::D1Spacing](#) to specify spacing between pattern instances.
- [ILocalLinearPatternFeatureData::D1TotalInstances](#) to specify the number of pattern instances.
- [ILocalLinearPatternFeatureData::D1ReverseDirection](#) to reverse Direction 1.
- [ILocalLinearPatternFeatureData::RotateInstances](#) to rotate pattern instances. If true:

- Use [ILocalLinearPatternFeatureData::RotationAngle](#).
- Use [ILocalLinearPatternFeatureData::RotationAxis](#).
- Use [ILocalLinearPatternFeatureData::FixedAxisOfRotation](#). If true: Use [ILocalLinearPatternFeatureData::AlignToSeed](#). If true: Use [ILocalLinearPatternFeatureData::SeedAlignmentReferencePoint](#).

If offsetting from an up-to reference, use:

- [ILocalLinearPatternFeatureData::D1EndRefOffset](#) to specify the distance between the last pattern instance and the up-to reference.
- [ILocalLinearPatternFeatureData::D1EndRefReverseOffset](#) to reverse the offset.
- [ILocalLinearPatternFeatureData::D1EndUseSpacing](#) to specify either spacing or number of instances (one or the other, not both):
 - If true, use [ILocalLinearPatternFeatureData::D1Spacing](#).
 - If false, use [ILocalLinearPatternFeatureData::D1TotalInstances](#).
- All of the [ILocalLinearPatternFeatureData::D2*](#) properties to specify Direction 2 for a bidirectional linear component pattern.
- Other properties such as [instances to skip](#) and [synchronize movement of flexible subassembly components](#).

You must call [IFeatureManager::CreateFeature](#) after initializing this feature data object in order to successfully create the linear pattern feature.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

 Accessors

[IFeature::GetDefinition](#)

IFeatureManager::CreateDefinition

 Access Diagram

[LocalLinearPatternFeatureData](#)

 See Also

[ILocalLinearPatternFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILinearPatternFeatureData Interface](#)

ILocalLinearPatternFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ILocalLinearPatternFeatureData Interface

For a list of all members of this type, see [ILocalLinearPatternFeatureData members](#).

 Public Methods

Name	Description
 AccessSelections	Gains access to selections that define this linear component pattern feature.
 GetD1AxisType	Gets the type of axis used for Direction 1 in this linear component pattern feature.
 GetD2AxisType	Gets the type of axis used for Direction 2 in this linear component pattern feature.
 GetModifiedInstance	Gets the modified values for the specified pattern instance in this linear component pattern feature.
 GetModifiedInstances	Gets the instance numbers of all modified instances in this linear component pattern.
 GetSeedComponentCount	Gets the number of seed components in this linear component pattern feature.
 GetSkippedItemCount	Gets the number of items that are skipped in this linear component pattern feature.
 GetTransform	Gets the transform for the specified instance of this linear component pattern feature.
 IAccessSelections	Obsolete. Superseded by ILocalLinearPatternFeatureData.IAccessSelections2.
 IAccessSelections2	Allows you to gain access to the selections that describe this linear component pattern feature.
 IGetSeedComponentArray	Gets an array of seed components for this linear component pattern feature.
 IGetSkippedItemArray	Gets the skipped components in this linear component pattern feature.
 ISetSeedComponentArray	Sets an array of seed component features to pattern for this linear component pattern feature.
 ISetSkippedItemArray	Sets the skipped components in this linear component pattern feature.
 ModifyInstance	Modifies the specified pattern instance with the specified distances in Directions 1 and 2 in this linear component pattern feature.
 ReleaseSelectionAccess	Releases access to the selections that created this linear component pattern feature.

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 See Also

[ILocalLinearPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILinearPatternFeatureData Interface](#)

ILocalLinearPatternFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



The following tables list the members exposed by [ILocalLinearPatternFeatureData](#).

Public Properties

Name	Description
 AlignToSeed	Gets or sets whether to align each instance to match the original alignment of the seed component in this linear component pattern feature.
 D1Axis	Gets or sets Direction 1 for this linear component pattern feature.
 D1EndCondition	Gets or sets how to specify the spacing of pattern instances in Direction 1 of this linear component pattern feature.
 D1EndReference	Gets or sets the up-to reference entity in Direction 1 for this linear component pattern feature.
 D1EndReferenceType	Gets the type of ILocalLinearPatternFeatureData.D1EndReference .
 D1EndRefOffset	Gets or sets the offset from a reference entity in Direction 1 of this linear component pattern feature.
 D1EndRefReverseOffset	Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 1 of this linear component pattern feature.
 D1EndSeedReference	Gets or sets the seed feature geometry to offset from the up-to reference geometry in Direction 1 of this linear component pattern feature.
 D1EndSeedReferenceType	Gets the type of ILocalLinearPatternFeatureData.D1EndSeedReference .
 D1EndUseSeedReference	Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 1 of this linear component pattern feature.
 D1EndUseSpacing	Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 1 for this linear component pattern feature.
 D1ReverseDirection	Gets or sets whether to reverse Direction 1 in this linear component pattern feature.
 D1Spacing	Gets the distance between pattern instances in Direction 1 of this linear component pattern feature.
 D1TotalInstances	Gets or sets the total number of instances in Direction 1, including skipped items, in this linear component pattern feature.
 D2Axis	Gets or sets Direction 2 for this bidirectional linear component pattern feature.
 D2EndCondition	Gets or sets how to specify the spacing of pattern instances in Direction 2 of this bidirectional linear component pattern feature.
 D2EndReference	Gets or sets the up-to reference geometry in Direction 2 for this bidirectional linear component pattern feature.
 D2EndReferenceType	Gets the type ILocalLinearPatternFeatureData.D2EndReference .
 D2EndRefOffset	Gets or sets the distance of the last pattern instance from up-to reference geometry in Direction 2 of this bidirectional linear component pattern feature.
 D2EndRefReverseOffset	Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 2 of this bidirectional linear component pattern feature.
 D2EndSeedReference	Gets or sets the seed feature geometry to offset from the up-to reference geometry in Direction 2 of this bidirectional linear component pattern feature.
 D2EndSeedReferenceType	Gets the type of ILocalLinearPatternFeatureData.D2EndSeedReference .
 D2EndUseSeedReference	Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 2 of this bidirectional linear component pattern feature.
 D2EndUseSpacing	Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 2 for this bidirectional linear component pattern feature.
 D2PatternSeedOnly	Gets or sets whether to create a pattern in Direction 2 using the seed component only, without replicating the pattern instances of Direction 1, in this bidirectional linear component pattern feature.
 D2ReverseDirection	Gets or sets whether to reverse Direction 2 in this bidirectional linear component pattern feature.
 D2Spacing	Gets the spacing between pattern instances in Direction 2 in this bidirectional linear component pattern feature.
 D2TotalInstances	Gets or sets the total number of instances in Direction 2, including skipped items, in this bidirectional linear component pattern feature.
 FixedAxisOfRotation	Gets or sets whether patterned instances rotate around a common axis.
 ForceUseSeedConfiguration	Gets or sets whether to synchronize the configuration of pattern components with the configuration of the seed component in this local linear pattern feature.
 ReverseAxisOfRotation	Gets or sets whether to reverse the direction of rotation.
 RotateInstances	Gets or sets whether to rotate components in Direction 1 of this linear component pattern.
 RotationAngle	Gets or sets the angle of rotation of instances in this linear component pattern.
 RotationAxis	Gets or sets the axis of rotation of components in this linear component pattern feature.
 SeedAlignmentReferencePoint	Gets or sets the type of reference point with which to align each pattern instance to the seed feature.
 SeedComponentArray	Gets or sets the seed components for this linear component pattern feature.
 SkippedItemArray	Gets or sets the skipped components in this linear component pattern feature.
 SynchronizeFlexibleComponents	Gets or sets whether to synchronize the movement of patterned flexible subassembly components with seed flexible subassembly components in this linear component pattern feature.

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Public Methods

Name	Description
 AccessSelections	Gains access to selections that define this linear component pattern feature.
 GetD1AxisType	Gets the type of axis used for Direction 1 in this linear component pattern feature.
 GetD2AxisType	Gets the type of axis used for Direction 2 in this linear component pattern feature.
 GetModifiedInstance	Gets the modified values for the specified pattern instance in this linear component pattern feature.
 GetModifiedInstances	Gets the instance numbers of all modified instances in this linear component pattern.
 GetSeedComponentCount	Gets the number of seed components in this linear component pattern feature.
 GetSkippedItemCount	Gets the number of items that are skipped in this linear component pattern feature.
 GetTransform	Gets the transform for the specified instance of this linear component pattern feature.
 IAccessSelections	Obsolete. Superseded by ILocalLinearPatternFeatureData.IAccessSelections2
 IAccessSelections2	Allows you to gain access to the selections that describe this linear component pattern feature.
 IGetSeedComponentArray	Gets an array of seed components for this linear component pattern feature.
 IGetSkippedItemArray	Gets the skipped components in this linear component pattern feature.
 ISetSeedComponentArray	Sets an array of seed component features to pattern for this linear component pattern feature.
 ISetSkippedItemArray	Sets the skipped components in this linear component pattern feature.
 ModifyInstance	Modifies the specified pattern instance with the specified distances in Directions 1 and 2 in this linear component pattern feature.
 ReleaseSelectionAccess	Releases access to the selections that created this linear component pattern feature.

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See Also

[ILocalLinearPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
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ILocalLinearPatternFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ILocalLinearPatternFeatureData Interface

For a list of all members of this type, see [ILocalLinearPatternFeatureData members](#).

Public Properties

Name	Description
 AlignToSeed	Gets or sets whether to align each instance to match the original alignment of the seed component in this linear component pattern feature.
 D1Axis	Gets or sets Direction 1 for this linear component pattern feature.
 D1EndCondition	Gets or sets how to specify the spacing of pattern instances in Direction 1 of this linear component pattern feature.
 D1EndReference	Gets or sets the up-to reference entity in Direction 1 for this linear component pattern feature.
 D1EndReferenceType	Gets the type of ILocalLinearPatternFeatureData.D1EndReference .
 D1EndRefOffset	Gets or sets the offset from a reference entity in Direction 1 of this linear component pattern feature.
 D1EndRefReverseOffset	Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 1 of this linear component pattern feature.
 D1EndSeedReference	Gets or sets the seed feature geometry to offset from the up-to-reference geometry in Direction 1 of this linear componnet pattern feature.
 D1EndSeedReferenceType	Gets the type of ILocalLinearPatternFeatureData.D1EndSeedReference .
 D1EndUseSeedReference	Gets or sets whether to offset a pattern seed reference or a centroid from the up-to-reference geometry in Direction 1 of this linear component pattern feature.
 D1EndUseSpacing	Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to-reference geometry in Direction 1 for this linear component pattern feature.
 D1ReverseDirection	Gets or sets whether to reverse Direction 1 in this linear component pattern feature.

	D1Spacing	Gets the distance between pattern instances in Direction 1 of this linear component pattern feature.
	D1TotalInstances	Gets or sets the total number of instances in Direction 1, including skipped items, in this linear component pattern feature.
	D2Axis	Gets or sets Direction 2 for this bidirectional linear component pattern feature.
	D2EndCondition	Gets or sets how to specify the spacing of pattern instances in Direction 2 of this bidirectional linear component pattern feature.
	D2EndReference	Gets or sets the up-to reference geometry in Direction 2 for this bidirectional linear component pattern feature.
	D2EndReferenceType	Gets the type of ILocalLinearPatternFeatureData.D2EndReference .
	D2EndRefOffset	Gets or sets the distance of the last pattern instance from up-to reference geometry in Direction 2 of this bidirectional linear component pattern feature.
	D2EndRefReverseOffset	Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 2 of this bidirectional linear component pattern feature.
	D2EndSeedReference	Gets or sets the seed feature geometry to offset from the up-to reference geometry in Direction 2 of this bidirectional linear component pattern feature.
	D2EndSeedReferenceType	Gets the type of ILocalLinearPatternFeatureData.D2EndSeedReference .
	D2EndUseSeedReference	Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 2 of this bidirectional linear component pattern feature.
	D2EndUseSpacing	Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 2 for this bidirectional linear component pattern feature.
	D2PatternSeedOnly	Gets or sets whether to create a pattern in Direction 2 using the seed component only, without replicating the pattern instances of Direction 1, in this bidirectional linear component pattern feature.
	D2ReverseDirection	Gets or sets whether to reverse Direction 2 in this bidirectional linear component pattern feature.
	D2Spacing	Gets the spacing between pattern instances in Direction 2 in this bidirectional linear component pattern feature.
	D2TotalInstances	Gets or sets the total number of instances in Direction 2, including skipped items, in this bidirectional linear component pattern feature.
	FixedAxisOfRotation	Gets or sets whether patterned instances rotate around a common axis.
	ForceUseSeedConfiguration	Gets or sets whether to synchronize the configuration of pattern components with the configuration of the seed component in this local linear pattern feature.
	ReverseAxisOfRotation	Gets or sets whether to reverse the direction of rotation.
	RotateInstances	Gets or sets whether to rotate components in Direction 1 of this linear component pattern.
	RotationAngle	Gets or sets the angle of rotation of instances in this linear component pattern.
	RotationAxis	Gets or sets the axis of rotation of components in this linear component pattern feature.
	SeedAlignmentReferencePoint	Gets or sets the type of reference point with which to align each pattern instance to the seed feature.
	SeedComponentArray	Gets or sets the seed components for this linear component pattern feature.
	SkippedItemArray	Gets or sets the skipped components in this linear component pattern feature.
	SynchronizeFlexibleComponents	Gets or sets whether to synchronize the movement of patterned flexible subassembly components with seed flexible subassembly components in this linear component pattern feature.

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 See Also

[ILocalLinearPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
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AccessSelections Method (ILocalLinearPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : AccessSelections Method (ILocalLinearPatternFeatureData)

Gains access to selections that define this linear component pattern feature.

 .NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections( _  
    ByVal TopDoc As System.Object, _  
    ByVal Component As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim TopDoc As System.Object
Dim Component As System.Object
Dim value As System.Boolean

value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(
    System.object TopDoc,
    System.object Component
)
```

C++/CLI

```
System.bool AccessSelections(
    & System.Object^ TopDoc,
    & System.Object^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc
Top-level document (see **Remarks**)

Component
Component in which the feature is to be modified (see **Remarks**)

Return Value

True if the selections were successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::AccessSelections](#).

Example

[Get and Set Seed Components \(C#\)](#)
[Get and Set Seed Components \(VB.NET\)](#)
[Get and Set Seed Components \(VBA\)](#)

Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) if you modified the feature
- [ILocalLinearPatternFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::IAccessSelections2 Method](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

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AlignToSeed Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : AlignToSeed Property (ILocalLinearPatternFeatureData)

Gets or sets whether to align each instance to match the original alignment of the seed component in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property AlignToSeed As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Boolean

instance.AlignToSeed = value

value = instance.AlignToSeed
```

C#

```
System.bool AlignToSeed {get; set;}
```

C++/CLI

```
property System.bool AlignToSeed {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to align each instance to the seed component, false to not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::AlignToSeed](#).

Remarks

This property is valid only if:

- [ILocalLinearPatternFeatureData::RotateInstances](#) is true,
- and -
- [ILocalLinearPatternFeatureData::FixedAxisOfRotation](#) is true.

If this property is set to true, you can specify how to align pattern instances to the seed component using [ILocalLinearPatternFeatureData::SeedAlignmentReferencePoint](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1Axis Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1Axis Property (ILocalLinearPatternFeatureData)

Gets or sets Direction 1 for this linear component pattern feature.

.NET Syntax

Property D1Axis As System.Object



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Object

instance.D1Axis = value

value = instance.D1Axis
```

```
System.Object D1Axis {get; set;}
```

```
property System.Object^ D1Axis {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Direction 1 entity: linear [edge](#), [line](#), [axis](#), [dimension](#), planar [face](#), planar [surface](#), conical face, conical surface, circular edge, or reference [plane](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1Axis](#).

Remarks

Use [ILocalLinearPatternFeatureData::GetD1AxisType](#) to determine the type of object to pass to this property.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::D1ReverseDirection Property](#)
[ILocalLinearPatternFeatureData::D1Spacing Property](#)
[ILocalLinearPatternFeatureData::D1TotalInstances Property](#)
[ILocalLinearPatternFeatureData::GetD1AxisType Method](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

D1EndCondition Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1EndCondition Property (ILocalLinearPatternFeatureData)

Gets or sets how to specify the spacing of pattern instances in Direction 1 of this linear component pattern feature.

.NET Syntax

Property D1EndCondition As System.Integer



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Integer
```

```
instance.D1EndCondition = value  
value = instance.D1EndCondition
```

C#

```
System.int D1EndCondition {get; set;}
```

C++/CLI

```
property System.int D1EndCondition {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Spacing of pattern instances in Direction 1 as defined in [swPatternEndCondition_e](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1EndCondition](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::RotateInstances Property \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

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D1EndReference Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1EndReference Property (ILocalLinearPatternFeatureData)

Gets or sets the up-to reference entity in Direction 1 for this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D1EndReference As System.Object
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Object
```

```
instance.D1EndReference = value
```

```
value = instance.D1EndReference
```

C#

```
System.object D1EndReference {get; set;}
```

C++/CLI

```
property System.Object^ D1EndReference {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pattern's up-to reference ([vertex](#), [edge](#), [face](#), or [plane](#)) in Direction 1

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1EndReference](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

To pre-select, use selection Mark = 256.

During edit of this property, if you specify an invalid entity [IFeature::ModifyDefinition](#) returns false and the feature does not roll back. To roll back the feature, call [IModelDoc2::ForceRebuild3](#) or move the rollback bar in the user interface to before and after the feature.

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::D1EndReferenceType Property \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1EndReferenceType Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1EndReferenceType Property (ILocalLinearPatternFeatureData)

Gets the type of [ILocalLinearPatternFeatureData::D1EndReference](#).

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property D1EndReferenceType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.D1EndReferenceType
```

C#

```
System.int D1EndReferenceType {get;}
```

C++/CLI

```
property System.int D1EndReferenceType {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type as defined in [swSelectType_e](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1EndReferenceType](#).

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1EndRefOffset Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1EndRefOffset Property (ILocalLinearPatternFeatureData)

Gets or sets the offset from a reference entity in Direction 1 of this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1EndRefOffset As System.Double

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Double
```

```
instance.D1EndRefOffset = value
```

```
value = instance.D1EndRefOffset
```

C#

```
System.double D1EndRefOffset {get; set;}
```

C++/CLI

```
property System.double D1EndRefOffset {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Offset of a pattern instance or centroid from [ILocalLinearPatternFeatureData::D1EndReference](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1EndRefOffset](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

The pattern instance or centroid to offset is:

- Specified by [ILocalLinearPatternFeatureData::D1EndSeedReference](#).
- Governed by [ILocalLinearPatternFeatureData::D1EndUseSeedReference](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1EndRefReverseOffset Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1EndRefReverseOffset Property (ILocalLinearPatternFeatureData)

Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 1 of this linear component pattern feature.

INET Syntax

Visual Basic (Declaration)

Property D1EndRefReverseOffset As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D1EndRefReverseOffset = value  
value = instance.D1EndRefReverseOffset
```

C#

```
System.bool D1EndRefReverseOffset {get; set;}
```

C++/CLI

```
property System.bool D1EndRefReverseOffset {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse [ILocalLinearPatternFeatureData::D1EndRefOffset](#) from [ILocalLinearPatternFeatureData::D1EndReference](#), false to not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1EndRefReverseOffset](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

D1EndSeedReference Property (ILocalLinearPatternFeatureData)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1EndSeedReference Property (ILocalLinearPatternFeatureData)

Gets or sets the seed feature geometry to offset from the up-to reference geometry in Direction 1 of this linear componnet pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1EndSeedReference As System.Object

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Object
```

```
instance.D1EndSeedReference = value
```

```
value = instance.D1EndSeedReference
```

C#

```
System.object D1EndSeedReference {get; set;}
```

C++/CLI

```
property System.Object^ D1EndSeedReference {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Seed geometry ([vertex](#), [edge](#), [face](#), or [plane](#)) to offset from [ILocalLinearPatternFeatureData::D1EndReference](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1EndSeedReference](#).

Remarks

This property is valid only if:

- [ILocalLinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#)
- and -
- [ILocalLinearPatternFeatureData::D1EndUseSeedReference](#) is true.

See [Accessing Selections that Define Features](#) for additional details on using this property.For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::D1EndSeedReferenceType Property \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

D1EndSeedReferenceType Property (ILocalLinearPatternFeatureData)

[See Also](#)

Gets the type of [ILocalLinearPatternFeatureData::D1EndSeedReference](#).

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property D1EndSeedReferenceType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer  
  
value = instance.D1EndSeedReferenceType
```

C#

```
System.int D1EndSeedReferenceType {get;}
```

C++/CLI

```
property System.int D1EndSeedReferenceType {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type as defined in [swSelectType_e](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1EndSeedReferenceType](#).

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1EndUseSeedReference Property (ILocalLinearPatternFeatureData)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [ILocalLinearPatternFeatureData Interface](#) : D1EndUseSeedReference Property (ILocalLinearPatternFeatureData)

Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 1 of this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D1EndUseSeedReference As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Boolean  
  
instance.D1EndUseSeedReference = value  
  
value = instance.D1EndUseSeedReference
```

C#

```
System.bool D1EndUseSeedReference {get; set;}
```

C++/CLI

```
property System.bool D1EndUseSeedReference {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to offset a pattern seed reference, false to offset a centroid

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1EndUseSeedReference](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

To pre-select, use selection Mark = 512.

If this property is set to true, use [ILocalLinearPatternFeatureData::D1EndSeedReference](#) to specify the seed geometry to offset from [ILocalLinearPatternFeatureData::D1EndReference](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1EndUseSpacing Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1EndUseSpacing Property (ILocalLinearPatternFeatureData)

Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 1 for this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D1EndUseSpacing As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D1EndUseSpacing = value
```

```
value = instance.D1EndUseSpacing
```

C#

```
System.bool D1EndUseSpacing {get; set;}
```

C++/CLI

```
property System.bool D1EndUseSpacing {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use spacing, false to use number of pattern instances in Direction 1

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1EndUseSpacing](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1ReverseDirection Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1ReverseDirection Property (ILocalLinearPatternFeatureData)

Gets or sets whether to reverse Direction 1 in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D1ReverseDirection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Boolean
```

```
instance.D1ReverseDirection = value
```

```
value = instance.D1ReverseDirection
```

C#

```
System.bool D1ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool D1ReverseDirection {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse Direction 1, false to not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1ReverseDirection](#).

Example

See the [ILocalLinearPatternFeatureData](#) example.

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_SpacingAndInstances](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetD1AxisType Method](#)
[ILocalLinearPatternFeatureData::D1Axis Property](#)
[ILocalLinearPatternFeatureData::D1Spacing Property](#)
[ILocalLinearPatternFeatureData::D1TotalInstances Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1Spacing Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1Spacing Property (ILocalLinearPatternFeatureData)

Gets the distance between pattern instances in Direction 1 of this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1Spacing As System.Double

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Double
```

```
instance.D1Spacing = value
```

```
value = instance.D1Spacing
```

C#

```
System.double D1Spacing {get; set;}
```

C++/CLI

```
property System.double D1Spacing {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance in meters between pattern instances in Direction 1

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1Spacing](#).

Example

See the [ILocalLinearPatternFeatureData](#) example.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetD1AxisType Method](#)
[ILocalLinearPatternFeatureData::D1Axis Property](#)
[ILocalLinearPatternFeatureData::D1ReverseDirection Property](#)
[ILocalLinearPatternFeatureData::D1TotalInstances Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1TotalInstances Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D1TotalInstances Property (ILocalLinearPatternFeatureData)

Gets or sets the total number of instances in Direction 1, including skipped items, in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1TotalInstances As System.Integer

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer
```

```
instance.D1TotalInstances = value
```

```
value = instance.D1TotalInstances
```

C#

```
System.int D1TotalInstances {get; set;}
```

C++/CLI

```
property System.int D1TotalInstances {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Number of total instances in Direction 1

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D1TotalInstances](#).

Example

See the [ILocalLinearPatternFeatureData](#) example.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetD1AxisType Method](#)
[ILocalLinearPatternFeatureData::D1Axis Property](#)
[ILocalLinearPatternFeatureData::D1ReverseDirection Property](#)
[ILocalLinearPatternFeatureData::D1Spacing Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2Axis Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D2Axis Property (ILocalLinearPatternFeatureData)

Gets or sets Direction 2 for this bidirectional linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2Axis As System.Object

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Object
```

```
instance.D2Axis = value
```

```
value = instance.D2Axis
```

C#

```
System.Object D2Axis {get; set;}
```

C++/CLI

```
property System.Object^ D2Axis {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Direction 2 entity: linear [edge](#), [line](#), [axis](#), [dimension](#), planar [face](#), planar [surface](#), conical face, conical surface, circular edge, or reference [plane](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2Axis](#).

Remarks

Use [ILocalLinearPatternFeatureData::GetD2AxisType](#) to determine the type of object to pass to this property.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::D2ReverseDirection Property](#)
[ILocalLinearPatternFeatureData::D2Spacing Property](#)
[ILocalLinearPatternFeatureData::D2TotalInstances Property](#)
[ILocalLinearPatternFeatureData::GetD2AxisType Method](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2EndCondition Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D2EndCondition Property (ILocalLinearPatternFeatureData)

Gets or sets how to specify the spacing of pattern instances in Direction 2 of this bidirectional linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2EndCondition As System.Integer

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer
```

```
instance.D2EndCondition = value
```

```
value = instance.D2EndCondition
```

C#

```
System.int D2EndCondition {get; set;}
```

C++/CLI

```
property System.int D2EndCondition {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Spacing of pattern instances in Direction 2 as defined in [swPatternEndCondition_e](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2EndCondition](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

D2EndReference Property (ILocalLinearPatternFeatureData)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D2EndReference Property (ILocalLinearPatternFeatureData)

Gets or sets the up-to reference geometry in Direction 2 for this bidirectional linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2EndReference As System.Object

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Object

instance.D2EndReference = value
value = instance.D2EndReference
```

C#

```
System.Object D2EndReference {get; set;}
```

C++/CLI

```
property System.Object^ D2EndReference {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).**Property Value**Pattern's up-to reference ([vertex](#), [edge](#), [face](#), or [plane](#)) in Direction 2

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2EndReference](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

To pre-select, use selection Mark = 1024.

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::D2EndReferenceType Property\(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

D2EndReferenceType Property (ILocalLinearPatternFeatureData)

[See Also](#)

Gets the type [ILocalLinearPatternFeatureData::D2EndReference](#).

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property D2EndReferenceType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer  
  
value = instance.D2EndReferenceType
```

C#

```
System.int D2EndReferenceType {get;}
```

C++/CLI

```
property System.int D2EndReferenceType {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type as defined in [swSelectType_e](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2EndReferenceType](#).

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

D2EndRefOffset Property (ILocalLinearPatternFeatureData)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [ILocalLinearPatternFeatureData Interface](#) : D2EndRefOffset Property (ILocalLinearPatternFeatureData)

Gets or sets the distance of the last pattern instance from up-to reference geometry in Direction 2 of this bidirectional linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D2EndRefOffset As System.Double
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Double  
  
instance.D2EndRefOffset = value  
  
value = instance.D2EndRefOffset
```

C#

```
System.double D2EndRefOffset {get; set;}
```

C++/CLI

```
property System.double D2EndRefOffset {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Offset of the last pattern instance from [ILocalLinearPatternFeatureData::D2EndReference](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2EndRefOffset](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2EndRefReverseOffset Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D2EndRefReverseOffset Property (ILocalLinearPatternFeatureData)

Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 2 of this bidirectional linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D2EndRefReverseOffset As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D2EndRefReverseOffset = value  
value = instance.D2EndRefReverseOffset
```

C#

```
System.bool D2EndRefReverseOffset {get; set;}
```

C++/CLI

```
property System.bool D2EndRefReverseOffset {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse [ILocalLinearPatternFeatureData::D2EndRefOffset](#) from [ILocalLinearPatternFeatureData::D2EndReference](#), false to not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2EndRefReverseOffset](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

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D2EndSeedReference Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D2EndSeedReference Property (ILocalLinearPatternFeatureData)

Gets or sets the seed feature geometry to offset from the up-to reference geometry in Direction 2 of this bidirectional linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D2EndSeedReference As System.Object
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Object
```

```
instance.D2EndSeedReference = value
```

```
value = instance.D2EndSeedReference
```

C#

```
System.object D2EndSeedReference {get; set;}
```

C++/CLI

```
property System.Object^ D2EndSeedReference {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Seed geometry ([vertex](#), [edge](#), [face](#), or [plane](#)) to offset from [ILocalLinearPatternFeatureData::D2EndReference](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2EndSeedReference](#).

Remarks

This property is valid only if:

- [ILocalLinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#),
- and -
- [ILocalLinearPatternFeatureData::D2EndUseSeedReference](#) is true.

To pre-select, use selection Mark = 2048.

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::D2EndSeedReferenceType Property \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2EndSeedReferenceType Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D2EndSeedReferenceType Property (ILocalLinearPatternFeatureData)

Gets the type of [ILocalLinearPatternFeatureData::D2EndSeedReference](#).

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property D2EndSeedReferenceType As System.Integer

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.D2EndSeedReferenceType
```

C#

```
System.int D2EndSeedReferenceType {get;}
```

C++/CLI

```
property System.int D2EndSeedReferenceType {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type as defined in [swSelectType_e](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2EndSeedReferenceType](#).

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2EndUseSeedReference Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D2EndUseSeedReference Property (ILocalLinearPatternFeatureData)

Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 2 of this bidirectional linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2EndUseSeedReference As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D2EndUseSeedReference = value
```

```
value = instance.D2EndUseSeedReference
```

C#

```
System.bool D2EndUseSeedReference {get; set;}
```

C++/CLI

```
property System.bool D2EndUseSeedReference {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to offset a pattern seed reference, false to offset a centroid

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2EndUseSeedReference](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

If this property is set to true, use [ILocalLinearPatternFeatureData::D2EndSeedReference](#) to specify the seed geometry to offset from [ILocalLinearPatternFeatureData::D2EndReference](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2EndUseSpacing Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D2EndUseSpacing Property (ILocalLinearPatternFeatureData)

Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 2 for this bidirectional linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2EndUseSpacing As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Boolean
```

```
instance.D2EndUseSpacing = value
```

```
value = instance.D2EndUseSpacing
```

C#

```
System.bool D2EndUseSpacing {get; set;}
```

C++/CLI

```
property System.bool D2EndUseSpacing {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use spacing, false to use number of pattern instances in Direction 2

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2EndUseSpacing](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2PatternSeedOnly Property (ILocalLinearPatternFeatureData)

[See Also](#)



Gets or sets whether to create a pattern in Direction 2 using the seed component only, without replicating the pattern instances of Direction 1, in this bidirectional linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D2PatternSeedOnly As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D2PatternSeedOnly = value
```

```
value = instance.D2PatternSeedOnly
```

C#

```
System.bool D2PatternSeedOnly {get; set;}
```

C++/CLI

```
property System.bool D2PatternSeedOnly {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to create a pattern in Direction 2 using the seed component only without replicating the pattern instances of Direction 1, false to not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData.D2PatternSeedOnly](#).

Example

See the [ILocalLinearPatternFeatureData](#) example.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

D2ReverseDirection Property (ILocalLinearPatternFeatureData)

[See Also](#)



Gets or sets whether to reverse Direction 2 in this bidirectional linear component pattern feature.

.NET Syntax

```
Property D2ReverseDirection As System.Boolean
```



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Boolean

instance.D2ReverseDirection = value

value = instance.D2ReverseDirection
```

```
System.bool D2ReverseDirection {get; set;}
```

```
property System.bool D2ReverseDirection {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse Direction 2, false to not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData.D2ReverseDirection](#).

Example

See the [ILocalLinearPatternFeatureData](#) example.

Remarks

This property is valid only if [ILocalLinearPatternFeatureData.D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_SpacingAndInstances](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetD2AxisType Method](#)
[ILocalLinearPatternFeatureData::D1Axis Property](#)
[ILocalLinearPatternFeatureData::D1Spacing Property](#)
[ILocalLinearPatternFeatureData::D2TotalInstances Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

D2Spacing Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D2Spacing Property (ILocalLinearPatternFeatureData)

Gets the spacing between pattern instances in Direction 2 in this bidirectional linear component pattern feature.

.NET Syntax

Property D2Spacing As System.Double

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Double

instance.D2Spacing = value

value = instance.D2Spacing
```

C#

```
System.double D2Spacing {get; set;}
```

C++/CLI

```
property System.double D2Spacing {
    System.double get();
    void set ( &    System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance in meters between pattern instances in Direction 2

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2Spacing](#).

Example

See the [ILocalLinearPatternFeatureData](#) example.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetD2AxisType Method](#)
[ILocalLinearPatternFeatureData::D2Axis Property](#)
[ILocalLinearPatternFeatureData::D2ReverseDirection Property](#)
[ILocalLinearPatternFeatureData::D2TotalInstances Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2TotalInstances Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : D2TotalInstances Property (ILocalLinearPatternFeatureData)

Gets or sets the total number of instances in Direction 2, including skipped items, in this bidirectional linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2TotalInstances As System.Integer

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Integer

instance.D2TotalInstances = value

value = instance.D2TotalInstances
```

C#

```
System.int D2TotalInstances {get; set;}
```

C++/CLI

```
property System.int D2TotalInstances {
    System.int get();
    void set ( &    System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Number of total instances in Direction 2

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::D2TotalInstances](#).

Example

See the [ILocalLinearPatternFeatureData](#) example.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetD2AxisType Method](#)
[ILocalLinearPatternFeatureData::D2Axis Property](#)
[ILocalLinearPatternFeatureData::D2ReverseDirection Property](#)
[ILocalLinearPatternFeatureData::D2Spacing Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

FixedAxisOfRotation Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : FixedAxisOfRotation Property (ILocalLinearPatternFeatureData)

Gets or sets whether patterned instances rotate around a common axis.

.NET Syntax

Visual Basic (Declaration)

```
Property FixedAxisOfRotation As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Boolean

instance.FixedAxisOfRotation = value
```

```
value = instance.FixedAxisOfRotation
```

C#

```
System.bool FixedAxisOfRotation {get; set;}
```

C++/CLI

```
property System.bool FixedAxisOfRotation {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if patterned instances rotate around [ILocalLinearPatternFeatureData::RotationAxis](#), false if each instance rotates about its own axis

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::FixedAxisOfRotation](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::RotateInstances](#) is true.

If this property is set to:

- True, then you can also specify [ILocalLinearPatternFeatureData::AlignToSeed](#).
- False, then the rotation axis of each instance of a component is translated along Direction 1 and then rotated following the axis of rotation for that component.

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ForceUseSeedConfiguration Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : ForceUseSeedConfiguration Property (ILocalLinearPatternFeatureData)

Gets or sets whether to synchronize the configuration of pattern components with the configuration of the seed component in this local linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property ForceUseSeedConfiguration As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.ForceUseSeedConfiguration = value
```

```
value = instance.ForceUseSeedConfiguration
```

C#

```
System.bool ForceUseSeedConfiguration {get; set;}
```

C++/CLI

```
property System.bool ForceUseSeedConfiguration {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use the seed component configuration for all pattern components, false to not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::ForceUseSeedConfiguration](#).

Remarks

If this property is set to true, the **Referenced configuration** section of the Component Properties dialog is disabled, and you cannot use [IComponent2::ReferencedConfiguration](#) to change the pattern instance configuration.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetD1AxisType Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : GetD1AxisType Method (ILocalLinearPatternFeatureData)

Gets the type of axis used for Direction 1 in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetD1AxisType() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer  
  
value = instance.GetD1AxisType()
```

C#

```
System.int GetD1AxisType()
```

C++/CLI

```
System.int GetD1AxisType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Type of axis as defined in [swPatternReferenceTypes_e](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::GetD1AxisType](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetD2AxisType Method](#)
[ILocalLinearPatternFeatureData::D1Axis Property](#)
[ILocalLinearPatternFeatureData::D1ReverseDirection Property](#)
[ILocalLinearPatternFeatureData::D1Spacing Property](#)
[ILocalLinearPatternFeatureData::D1TotalInstances Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetD2AxisType Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : GetD2AxisType Method (ILocalLinearPatternFeatureData)

Gets the type of axis used for Direction 2 in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetD2AxisType() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer  
  
value = instance.GetD2AxisType()
```

C#

```
System.int GetD2AxisType()
```

C++/CLI

```
System.int GetD2AxisType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Type of axis as defined in [swPatternReferenceTypes_e](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::GetD2AxisType](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetD1AxisType Method](#)
[ILocalLinearPatternFeatureData::D2Axis Property](#)
[ILocalLinearPatternFeatureData::D2ReverseDirection Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetModifiedInstance Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : GetModifiedInstance Method (ILocalLinearPatternFeatureData)

Gets the modified values for the specified pattern instance in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetModifiedInstance( _  
    ByVal Instance As System.Integer, _  
    ByRef D1Distance As System.Double, _  
    ByRef D1NominalVal As System.Double, _  
    ByRef D2Distance As System.Double, _  
    ByRef D2NominalVal As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim Instance As System.Integer  
Dim D1Distance As System.Double  
Dim D1NominalVal As System.Double  
Dim D2Distance As System.Double  
Dim D2NominalVal As System.Double  
  
instance.GetModifiedInstance(Instance, D1Distance, D1NominalVal, D2Distance, D2NominalVal)
```

C#

```
void GetModifiedInstance(  
    System.int Instance,  
    out System.double D1Distance,  
    out System.double D1NominalVal,  
    out System.double D2Distance,  
    out System.double D2NominalVal  
)
```

C++/CLI

```
void GetModifiedInstance(  
& System.int Instance,  
& [Out] System.double D1Distance,  
& [Out] System.double D1NominalVal,  
& [Out] System.double D2Distance,  
& [Out] System.double D2NominalVal  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance

Pattern instance number (see **Remarks**)

D1Distance

Distance from the pattern seed in Direction 1

D1NominalVal

Offset from the nominal position of Instance in Direction 1

D2Distance

Distance from the pattern seed in Direction 2

D2NominalVal

Offset from the nominal position of Instance in Direction 2

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::GetModifiedInstance](#).

Remarks

To specify Instance, you can use [ILocalLinearPatternFeatureData::GetModifiedInstances](#) to learn which pattern instances are modified.

To calculate Instance:

1. Ensure that [ILocalLinearPatternFeatureData::D2PatternSeedOnly](#) is set to false.
2. Calculate:

For a bi-directional pattern:

$$I = n2 * (j - 1) + (i - 1)$$

For a uni-directional pattern:

$$I = i - 1$$

where:

- I = pattern instance number
- $n2$ = number of instances in Direction 2
- i = index for Direction 1
- j = index for Direction 2

In the pattern's PropertyManager, find $n2$ in the **Direction 2 > Spacing and Instances > Number of instances** field and find $[i,j]$ in the **Modified Instances** section.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

SOLIDWORKS API Help

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GetModifiedInstances Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : GetModifiedInstances Method (ILocalLinearPatternFeatureData)

Gets the instance numbers of all modified instances in this linear component pattern.

.NET Syntax

Visual Basic (Declaration)

```
Function GetModifiedInstances() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Object  
  
value = instance.GetModifiedInstances()
```

C#

```
System.object GetModifiedInstances()
```

C++/CLI

```
System.Object^ GetModifiedInstances();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of modified pattern instance numbers

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::GetModifiedInstances](#).

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetModifiedInstance Method\(\)](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSeedComponentCount Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : GetSeedComponentCount Method (ILocalLinearPatternFeatureData)

Gets the number of seed components in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSeedComponentCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer  
  
value = instance.GetSeedComponentCount()
```

C#

```
System.int GetSeedComponentCount()
```

C++/CLI

```
System.int GetSeedComponentCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of seed components

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::GetSeedComponentCount](#).

Remarks

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::SetSeedComponentArray Method](#)
[ILocalLinearPatternFeatureData::SeedComponentArray Property](#)
[ILocalLinearPatternFeatureData::GetSeedComponentArray Method](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSkippedItemCount Method (ILocalLinearPatternFeatureData)

[See Also](#)



Gets the number of items that are skipped in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSkippedItemCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer  
  
value = instance.GetSkippedItemCount()
```

C#

```
System.int GetSkippedItemCount()
```

C++/CLI

```
System.int GetSkippedItemCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of skipped items

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::GetSkippedItemCount](#).

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::IGetSkippedItemArray Method](#)
[ILocalLinearPatternFeatureData::ISetSkippedItemArray Method](#)
[ILocalLinearPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTransform Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : GetTransform Method (ILocalLinearPatternFeatureData)

Gets the transform for the specified instance of this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTransform( _  
    ByVal Instance As System.Integer _  
) As MathTransform
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim Instance As System.Integer  
Dim value As MathTransform  
  
value = instance.GetTransform(Instance)
```

C#

```
MathTransform GetTransform(  
    System.int Instance  
)
```

C++/CLI

```
MathTransform^ GetTransform(  
& System.int Instance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance

Index of one repeating element in this pattern (see **Remarks**)

Return Value

[Transform](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::GetTransform](#).

Remarks

1 <= *Instance* <= ([ILocalLinearPatternFeatureData::D1TotalInstances](#) + [ILocalLinearPatternFeatureData::D2TotalInstances](#))

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic](#).

IAccessSelections Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : IAccessSelections Method (ILocalLinearPatternFeatureData)

Obsolete. Superseded by [ILocalLinearPatternFeatureData::IAccessSelections2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IAccessSelections( _  
    ByVal TopDoc As ModelDoc, _  
    ByVal Component As Component _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim TopDoc As ModelDoc  
Dim Component As Component  
Dim value As System.Boolean  
  
value = instance.IAccessSelections(TopDoc, Component)
```

C#

```
System.bool IAccessSelections(  
    ModelDoc TopDoc,
```

```
        Component Component
    )
```

C++/CLI

```
System.bool IAccessSelections(
& ModelDoc^ TopDoc,
& Component^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Component

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::IAccessSelections](#).

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IAccessSelections2 Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : IAccessSelections2 Method (ILocalLinearPatternFeatureData)

Allows you to gain access to the selections that describe this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IAccessSelections2( _
    ByVal TopDoc As ModelDoc2, _
    ByVal Component As Component2 _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim TopDoc As ModelDoc2
Dim Component As Component2
Dim value As System.Boolean

value = instance.IAccessSelections2(TopDoc, Component)
```

C#

```
System.bool IAccessSelections2(
    ModelDoc2 TopDoc,
    Component2 Component
)
```

C++/CLI

```
System.bool IAccessSelections2(
& ModelDoc2^ TopDoc,
& Component2^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document (see **Remarks**)
Component
Component in which the feature is to be modified (see **Remarks**)

Return Value

True if the selections were successfully accessed, false if not

 Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::IAccessSelections2](#).

 Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::IModifyDefinition2](#) if you modified the feature
- [ILocalLinearPatternFeatureData::ReleaseSelectionAccess](#) if you did not

 See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::AccessSelections Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help [Send comments on this topic.](#)

IGetSeedComponentArray Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : IGetSeedComponentArray Method (ILocalLinearPatternFeatureData)

Gets an array of seed components for this linear component pattern feature.

 .NET Syntax

Visual Basic (Declaration)

Function IGetSeedComponentArray() As System.Object

Visual Basic (Usage)



Dim instance As [ILocalLinearPatternFeatureData](#)
Dim value As System.Object

value = instance.IGetSeedComponentArray()

C#

System.object IGetSeedComponentArray()

C++/CLI

System.Object^ IGetSeedComponentArray();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of seed component [features](#) (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

 Remarks

You can use [IFeature::GetSpecificFeature2](#) to get the [IComponent2](#) object for each seed component represented by a feature.
See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetSeedComponentCount Method](#)
[ILocalLinearPatternFeatureData::ISetSeedComponentArray Method](#)
[ILocalLinearPatternFeatureData::SeedComponentArray Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSkippedItemArray Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : IGetSkippedItemArray Method (ILocalLinearPatternFeatureData)

Gets the skipped components in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSkippedItemArray() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer  
  
value = instance.IGetSkippedItemArray()
```

C#

```
System.int IGetSkippedItemArray()
```

C++/CLI

```
System.int IGetSkippedItemArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of skipped pattern instances
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

This array is 0-based. If you skip the 3rd and 5th components in the pattern, then the array looks like ArrayOut[0] = 3 and ArrayOut[1] = 5.

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetSkippedItemCount Method](#)
[ILocalLinearPatternFeatureData::ISetSkippedItemArray Method](#)
[ILocalLinearPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

ISetSeedComponentArray Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : ISetSeedComponentArray Method (ILocalLinearPatternFeatureData)

Sets an array of seed component features to pattern for this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetSeedComponentArray( _
    ByVal FeatCount As System.Integer, _
    ByRef ArrayDataIn As System.Object _
)
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim FeatCount As System.Integer
Dim ArrayDataIn As System.Object

instance.ISetSeedComponentArray(FeatCount, ArrayDataIn)
```

C#

```
void ISetSeedComponentArray(
    System.int FeatCount,
    ref System.object ArrayDataIn
)
```

C++/CLI

```
void ISetSeedComponentArray(
    & System.int FeatCount,
    & System.Object^% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatCount

Number of seed components

ArrayDataIn

- In-process, unmanaged C++: Pointer to an array of seed component [features](#) (see **Remarks**)

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

You can pass a mixed array of [IComponent2](#) and [IFeature](#) objects representing components.

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetSeedComponentCount Method](#)
[ILocalLinearPatternFeatureData::ISetSeedComponentArray Method](#)
[ILocalLinearPatternFeatureData::SeedComponentArray Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

ISetSkippedItemArray Method (ILocalLinearPatternFeatureData)

[See Also](#)



Sets the skipped components in this linear component pattern feature.

VB.NET Syntax

Visual Basic (Declaration)

```
Sub ISetSkippedItemArray( _  
    ByVal FeatCount As System.Integer, _  
    ByRef ArrayDataIn As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim FeatCount As System.Integer  
Dim ArrayDataIn As System.Integer  
  
instance.ISetSkippedItemArray(FeatCount, ArrayDataIn)
```

C#

```
void ISetSkippedItemArray(  
    System.int FeatCount,  
    ref System.int ArrayDataIn  
)
```

C++/CLI

```
void ISetSkippedItemArray(  
& System.int FeatCount,  
& System.int% ArrayDataIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatCount

Number of components to skip

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of skipped components
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

ArrayDataIn is 0-based. If you skip the 3rd and 5th components in the pattern, the array looks like ArrayDataIn[0] = 3 and ArrayDataIn[1] = 5.

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILocalLinearPatternFeatureData Interface](#)

[ILocalLinearPatternFeatureData Members](#)

[ILocalLinearPatternFeatureData::GetSkippedItemCount Method](#)

[ILocalLinearPatternFeatureData::IGetSkippedItemArray Method](#)

[ILocalLinearPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2



Visual Basic Syntax

Visual Basic (Declaration)

```
Function ModifyInstance( _  
    ByVal Instance As System.Integer, _  
    ByVal D1Distance As System.Double, _  
    ByVal IsD1OffsetFromNominal As System.Boolean, _  
    ByVal D2Distance As System.Double, _  
    ByVal IsD2OffsetFromNominal As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim Instance As System.Integer  
Dim D1Distance As System.Double  
Dim IsD1OffsetFromNominal As System.Boolean  
Dim D2Distance As System.Double  
Dim IsD2OffsetFromNominal As System.Boolean  
Dim value As System.Boolean  
  
value = instance.ModifyInstance(Instance, D1Distance, IsD1OffsetFromNominal, D2Distance,  
IsD2OffsetFromNominal)
```

C#

```
System.bool ModifyInstance(  
    System.int Instance,  
    System.double D1Distance,  
    System.bool IsD1OffsetFromNominal,  
    System.double D2Distance,  
    System.bool IsD2OffsetFromNominal  
)
```

C++/CLI

```
System.bool ModifyInstance(  
& System.int Instance,  
& System.double D1Distance,  
& System.bool IsD1OffsetFromNominal,  
& System.double D2Distance,  
& System.bool IsD2OffsetFromNominal  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance

Pattern instance number (see **Remarks**)

D1Distance

Distance from the pattern seed in Direction 1

IsD1OffsetFromNominal

True if D1Distance is the offset from the nominal position of Instance, false if not

D2Distance

Distance from the pattern seed in Direction 2

IsD2OffsetFromNominal

True if D2Distance is the offset from the nominal position of Instance, false if not

Return Value

True if the pattern instance is modified successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::ModifyInstance](#).

Remarks

To calculate Instance:

1. Ensure that [LocalLinearPatternFeatureData::D2PatternSeedOnly](#) is set to false.
2. Calculate:

For a bi-directional pattern:

$$I = n2 * (j - 1) + (j - 1)$$

For a uni-directional pattern:

$$I = j - 1$$

where:

- I = pattern instance number
- $n2$ = number of instances in Direction 2
- i = index for Direction 1
- j = index for Direction 2

In the pattern's PropertyManager, find $n2$ in the **Direction 2 > Spacing and Instances > Number of instances** field and find $[i,j]$ in the **Modified Instances** section.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReleaseSelectionAccess Method (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : ReleaseSelectionAccess Method (ILocalLinearPatternFeatureData)

Releases access to the selections that created this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
```

```
instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::ReleaseSelectionAccess](#).

Remarks

[ILocalLinearPatternFeatureData::AccessSelections](#) and [ILocalLinearPatternFeatureData::!AccessSelections2](#) put the model into a rollback state to allow access to the selections that define this feature.

Use this method to restore the rollback state if you did not modify the feature or use [IFeature::ModifyDefinition](#) or [IFeature::!ModifyDefintion2](#) if you did.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReverseAxisOfRotation Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : ReverseAxisOfRotation Property (ILocalLinearPatternFeatureData)

Gets or sets whether to reverse the direction of rotation.

.NET Syntax

Visual Basic (Declaration)

Property ReverseAxisOfRotation As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.ReverseAxisOfRotation = value
```

```
value = instance.ReverseAxisOfRotation
```

C#

```
System.bool ReverseAxisOfRotation {get; set;}
```

C++/CLI

```
property System.bool ReverseAxisOfRotation {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse the direction of rotation, false to not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::ReverseAxisOfRotation](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::RotateInstances](#) is true.

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RotateInstances Property (ILocalLinearPatternFeatureData)

[See Also](#)



Gets or sets whether to rotate components in Direction 1 of this linear component pattern.

.NET Syntax

Visual Basic (Declaration)

```
Property RotateInstances As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Boolean
```

```
instance.RotateInstances = value
```

```
value = instance.RotateInstances
```

C#

```
System.bool RotateInstances {get; set;}
```

C++/CLI

```
property System.bool RotateInstances {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to rotate pattern instances, false to not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::RotateInstances](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_SpacingAndInstances](#).

If you set this property to true, you can also set:

- [ILocalLinearPatternFeatureData::RotationAngle](#)
- [ILocalLinearPatternFeatureData::RotationAxis](#)
- [ILocalLinearPatternFeatureData::ReverseAxisOfRotation](#)
- [ILocalLinearPatternFeatureData::FixedAxisOfRotation](#)

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

RotationAngle Property (ILocalLinearPatternFeatureData)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > ILocalLinearPatternFeatureData Interface : RotationAngle Property (ILocalLinearPatternFeatureData)

Gets or sets the angle of rotation of instances in this linear component pattern.

.NET Syntax

```
Property RotationAngle As System.Double
```



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Double

instance.RotationAngle = value
value = instance.RotationAngle
```

```
System.double RotationAngle {get; set;}
```

```
property System.double RotationAngle {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Angle of rotation in radians of pattern instances

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::RotationAngle](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::RotateInstances](#) is true.

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

RotationAxis Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : RotationAxis Property (ILocalLinearPatternFeatureData)

Gets or sets the axis of rotation of components in this linear component pattern feature.

.NET Syntax

```
Property RotationAxis As System.Object
```



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Object
```

```
instance.RotationAxis = value  
  
value = instance.RotationAxis
```

C#

```
System.Object RotationAxis {get; set;}
```

C++/CLI

```
property System.Object^ RotationAxis {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Axis](#), [line](#), or [edge](#) around which the pattern components rotate

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::RotationAxis](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::RotateInstances](#) is true.

The rotation axis must be parallel to [ILocalLinearPatternFeatureData::D1Axis](#).

Use:

- [ILocalLinearPatternFeatureData::FixedAxisOfRotation](#) to specify whether all instances of components rotate about this rotation axis.
- [ILocalLinearPatternFeatureData::ReverseAxisOfRotation](#) to reverse the direction of this rotation axis.

You cannot edit this property to null or Nothing after the feature is first created with a rotation axis. If you try, [IFeature::ModifyDefinition](#) returns false and the feature does not roll back. To roll back the feature, call [IModelDoc2::ForceRebuild3](#) or move the rollback bar in the user interface to before and after the feature.

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Component Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SeedAlignmentReferencePoint Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : SeedAlignmentReferencePoint Property (ILocalLinearPatternFeatureData)

Gets or sets the type of reference point with which to align each pattern instance to the seed feature.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property SeedAlignmentReferencePoint As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Integer  
  
instance.SeedAlignmentReferencePoint = value
```

```
value = instance.SeedAlignmentReferencePoint
```

C#

```
System.int SeedAlignmentReferencePoint {get; set;}
```

C++/CLI

```
property System.int SeedAlignmentReferencePoint {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Seed alignment reference point as defined in [swSeedAlignmentReferencePoint_e](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::SeedAlignmentReferencePoint](#).

Remarks

This property is valid only if [ILocalLinearPatternFeatureData::AlignToSeed](#) is true.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SeedComponentArray Property (ILocalLinearPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : SeedComponentArray Property (ILocalLinearPatternFeatureData)

Gets or sets the seed components for this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property SeedComponentArray As System.Object
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData  
Dim value As System.Object
```

```
instance.SeedComponentArray = value
```

```
value = instance.SeedComponentArray
```

C#

```
System.object SeedComponentArray {get; set;}
```

C++/CLI

```
property System.Object^ SeedComponentArray {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of seed component [features](#)

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::SeedComponentArray](#).

Example

[Get and Set Seed Components \(VBA\)](#)
[Get and Set Seed Components \(VB.NET\)](#)
[Get and Set Seed Components \(C#\)](#)

Remarks

You can:

- use [IFeature::GetSpecificFeature2](#) to get the [IComponent2](#) object for each seed component.
- pass a mixed array of [IComponent2](#) objects and [IFeature](#) objects representing components.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::GetSeedComponentCount Method](#)
[ILocalLinearPatternFeatureData::IGetSeedComponentArray Method](#)
[ILocalLinearPatternFeatureData::ISetSeedComponentArray Method](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

SkippedItemArray Property (ILocalLinearPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : SkippedItemArray Property (ILocalLinearPatternFeatureData)

Gets or sets the skipped components in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property SkippedItemArray As System.Object

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Object
```

```
instance.SkippedItemArray = value
```

```
value = instance.SkippedItemArray
```

C#

```
System.object SkippedItemArray {get; set;}
```

C++/CLI

```
property System.Object^ SkippedItemArray {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of integers representing the skipped pattern components

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::SkippedItemArray](#).

Example

[Get Number of Instances Skipped in Driving Feature \(C#\)](#)
[Get Number of Instances Skipped in Driving Feature \(VB.NET\)](#)
[Get Number of Instances Skipped in Driving Feature \(VBA\)](#)

Remarks

This array is 0-based. If you skip the third and fifth components, then the array looks like ArrayOut[0]=3 and ArrayOut[1]=5. To calculate the pattern instance numbers to skip:

Calculate:

For a bi-directional pattern:

$$I = n2 * (i - 1) + (j - 1)$$

For a uni-directional pattern:

$$I = i - 1$$

where:

- I = pattern instance number
- $n2$ = number of instances in Direction 2
- i = index for Direction 1
- j = index for Direction 2

In the pattern's PropertyManager, find $n2$ in the **Direction 2 > Spacing and Instances > Number of instances** field and find $[i,j]$ in the **Modified Instances** section.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)
[ILocalLinearPatternFeatureData::IGetSkippedItemCount Method](#)
[ILocalLinearPatternFeatureData::IGetSkippedItemArray Method](#)
[ILocalLinearPatternFeatureData::ISetSkippedItemArray Method](#)

Availability

SOLIDWORKS 2001 SP2, Revision Number 9.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

SynchronizeFlexibleComponents Property (ILocalLinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILocalLinearPatternFeatureData Interface](#) : SynchronizeFlexibleComponents Property (ILocalLinearPatternFeatureData)

Gets or sets whether to synchronize the movement of patterned flexible subassembly components with seed flexible subassembly components in this linear component pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property SynchronizeFlexibleComponents As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILocalLinearPatternFeatureData
Dim value As System.Boolean

instance.SynchronizeFlexibleComponents = value
value = instance.SynchronizeFlexibleComponents
```

C#

```
System.bool SynchronizeFlexibleComponents {get; set;}
```

C++/CLI

```
property System.bool SynchronizeFlexibleComponents {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to synchronize the movement of patterned flexible subassembly components with seed flexible subassembly components, false to not

Visual Basic for Applications (VBA) Syntax

See [LocalLinearPatternFeatureData::SynchronizeFlexibleComponents](#).

Example

See the [ILocalLinearPatternFeatureData](#) example.

Remarks

This property is valid for a linear pattern of flexible subassemblies.

If this property is set to true, then when you move components in the seed flexible subassembly, components in the patterned flexible subassemblies also move, and vice versa.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILocalLinearPatternFeatureData Interface](#)
[ILocalLinearPatternFeatureData Members](#)

Availability

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